



## LEVENT COLLEGE IB PROGRAMME ELECTRONICS & ROBOTICS CLUB ANNUAL CURRICULUM

|                            |                   |                      |   |
|----------------------------|-------------------|----------------------|---|
| Programme Name             | LEVENT COLLEGE IB | Instructor Name      |   |
| Programme Duration (weeks) | 40 weeks          | Participant Name     |   |
|                            |                   | Participant TR ID No | . |

| September 2024 |                                            |                                                           |
|----------------|--------------------------------------------|-----------------------------------------------------------|
| Week           | Subject                                    | Objective                                                 |
| 1. Week        | Project Introduction & Erasmus+ Briefing   | Understands the aim and scope of the project.             |
| 2. Week        | Drone Project Timeline & Task Distribution | Forms teamwork, shares roles, and creates a project plan. |
| 3. Week        | Flight Physics – Thrust, Weight, Direction | Recognizes the physical forces affecting drone flight.    |
| 4. Week        | Types of Multirotors                       | Explains multirotor types and their advantages.           |

| October 2024 |                                     |                                                            |
|--------------|-------------------------------------|------------------------------------------------------------|
| Week         | Subject                             | Objective                                                  |
| 5. Week      | Introduction to Electricity         | Explains the concepts of voltage, current, and resistance. |
| 6. Week      | Basic Circuit Setup                 | Reads and builds simple circuit diagrams.                  |
| 7. Week      | Brushless DC Motor (BLDC) Structure | Explains how motors work.                                  |
| 8. Week      | ESC Control & Connections           | Learns the function of an ESC and connects it correctly.   |

| November 2024 |                                    |                                                         |
|---------------|------------------------------------|---------------------------------------------------------|
| Week          | Subject                            | Objective                                               |
| 9. Week       | LiPo Battery Features              | Understands battery capacity, C-rating, and cell count. |
| 10. Week      | Battery Safety & Charging          | Applies safe usage and charging procedures.             |
| 11. Week      | Pixhawk / Flight Controller Boards | Identifies flight controller types and their functions. |
| 12. Week      | Loading Pixhawk Firmware           | Learns how to install PX4 / ArduPilot firmware.         |

| December 2024 |                                        |                                                           |
|---------------|----------------------------------------|-----------------------------------------------------------|
| Week          | Subject                                | Objective                                                 |
| 13. Week      | ESC Calibration                        | Performs ESC calibration using PWM signals.               |
| 14. Week      | PWM and Signal Logic                   | Understands PWM logic, frequency–duty cycle relationship. |
| 15. Week      | RC Transmitter – Receiver Introduction | Learns RC system and performs channel pairing.            |
| 16. Week      | RC Transmitter Testing                 | Monitors and analyzes channel data.                       |

| January 2025 |                               |                                                      |
|--------------|-------------------------------|------------------------------------------------------|
| Week         | Subject                       | Objective                                            |
| 17. Week     | GPS and GNSS Systems          | Learns the principle of positioning systems.         |
| 18. Week     | Connecting Pixhawk with GPS   | Connects GPS module properly to Pixhawk.             |
| 19. Week     | Drone Assembly – Beginning    | Starts assembling motor, ESC, and flight controller. |
| 20. Week     | Drone Assembly – Continuation | Completes wiring and connection checks.              |

| February 2025 |                                           |                                              |
|---------------|-------------------------------------------|----------------------------------------------|
| Week          | Subject                                   | Objective                                    |
| 21. Week      | QGroundControl Introduction               | Learns the ground control station interface. |
| 22. Week      | Flight Sensor Calibrations (IMU, Compass) | Performs sensor calibrations.                |
| 23. Week      | Flight Simulation                         | Tests basic flight scenarios in a simulator. |
| 24. Week      | Real-Time Flight Telemetry                | Learns to monitor real-time flight data.     |

| March 2025 |                                        |                                                      |
|------------|----------------------------------------|------------------------------------------------------|
| Week       | Subject                                | Objective                                            |
| 25. Week   | What is PID? – Fundamentals            | Understands PID control logic.                       |
| 26. Week   | PID Settings – Pitch/Roll/Yaw          | Performs basic PID adjustments.                      |
| 27. Week   | MAVLink Protocol Introduction          | Learns what MAVLink is and how it works.             |
| 28. Week   | Reading MAVLink Data with Raspberry Pi | Reads position data from Pixhawk using Raspberry Pi. |

| April 2025 |                              |                                                          |
|------------|------------------------------|----------------------------------------------------------|
| Week       | Subject                      | Objective                                                |
| 29. Week   | UBEC and 5V Power Supply     | Powers Raspberry Pi by stepping down high voltage to 5V. |
| 30. Week   | Installing Raspberry Pi      | Installs operating system and basic setup.               |
| 31. Week   | Connecting Pi Camera         | Connects camera and captures video.                      |
| 32. Week   | Capturing Images with Python | Processes camera images using OpenCV in Python.          |

| MAY 2025 |                                      |                                              |
|----------|--------------------------------------|----------------------------------------------|
| Week     | Subject                              | Objective                                    |
| 33. Week | Drone Test Flights                   | Performs test flights safely.                |
| 34. Week | Drone Safety Training                | Applies emergency procedures.                |
| 35. Week | Waypoint (Autonomous Route) Planning | Plans routes on a mission planner.           |
| 36. Week | Autonomous Flight Task               | Executes autonomous flight on planned route. |

| JUNE 2025 |                                  |                                                         |
|-----------|----------------------------------|---------------------------------------------------------|
| Week      | Subject                          | Objective                                               |
| 37. Week  | Drone Flight Logging & Reporting | Records and reports flight data.                        |
| 38. Week  | System Evaluation                | Identifies errors and shortcomings of the drone system. |
| 39. Week  | Presentation & Rehearsal         | Prepares and rehearses project presentation.            |
| 40. Week  | Final Demonstration & Exhibition | Presents drone system in front of an audience.          |